

The Fusion Nexus: Exploring the Confluence of Virtual and Real Worlds through Biocognitive Audio-Verbal Interface in Immersive XR Environments

Bowen Yuan
Bowen.yuan@mymail.unisa.edu.au
University of South Australia
Adelaide, Australia

Qianyuan Zou
qzou984@aucklanduni.ac.nz
The University of Auckland
Auckland, New Zealand

Gun Lee
Gun.Lee@unisa.edu.au
University of South Australia
Adelaide, Australia

Zirui Xiao
richardxiao@cmail.carleton.ca
Carleton University
Ottawa, Canada

Zhuang Chang
zcha621@aucklanduni.ac.nz
The University of Auckland
Auckland, New Zealand

Mark Billinghamurst
Mark.Billinghurst@unisa.edu.au
University of South Australia
Adelaide, Australia

Tamil Selvan Gunasekaran
themastergts007@gmail.com
The University of Auckland
Auckland, New Zealand

Misha Mirza
misha.mirza@pg.canterbury.ac.nz
HIT Lab NZ, University of Canterbury
Canterbury, New Zealand



Figure 1: System Set-up: a. Pose detection of Machine learning Algorithm; b. VR environment Overview; c. Image of Hear Rate data collected by Shimmer PPG Sensor; d. Intensity of drum beats played to the real-world player.

ABSTRACT

In this demo, we present FusionNexus, an asymmetric collaboration experience where a virtual-world player needs to guide a real-world player to make certain actions. A machine learning algorithm was used to do the pose detection of the real-world player. We collect the virtual-world player's heartbeat to control drum beats in background music played to the real-world player to convey emotional status of the virtual-world player. In this demonstration, two players will collaborate to accomplish a time task.

KEYWORDS

Cross reality, Communication, Asymmetric collaboration

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1 INTRODUCTION

Asymmetric collaboration between users in Virtual Reality (VR) and users in the real-world, a concept originating from virtual telepresence, is still relatively new [Steed et al. 2012]. Most research in this area focused on the collaboration between players observing the same environment from different perspectives [Chen et al. 2022]. Previous studies [Harris and Hancock 2019] [Chen et al. 2022] [Sugiura et al. 2018] found that in asymmetric collaborations, when

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players share or complement their perspectives, they are more inclined to communicate, share information, and express emotions. In contrast, in our demo "Fusion Nexus", We will explore how to convey the VR player's emotions and physical status in a different environment where the real-world player does not have the vision of the VR player and needs to follow the VR player's instructions.

The heart rate variability (HRV) has been proven to be an index of regulated emotional response [Quintana et al. 2012], and listeners have been able to perceive the intended emotions in music [Laukka and Gabrielsson 2000]. This inspired us to combine them. The primary innovation in our demo is that we collect the VR player's heartbeat and automatically generate different drum beats according to the change of its rate, trying to convey the emotion and physical status of the VR player to the real-world player.

2 SYSTEM OVERVIEW

When we designed the immersive environment of the demo, we aimed to continuously put pressure on the VR player through the environment, making them feel tense to trigger their changes in heart rate while playing. The scene is set in a garage that is flooded and the water level continues to rise. The real-world player needs to make certain poses to help the VR player survive. To determine whether the real-world player is performing the correct action, we employed a machine learning algorithm. The machine learning model was trained using a custom data-set, built upon the Movenet model[Votel and Na [n. d.]]. This data-set was generated by capturing participants performing a range of poses, in accordance with the system's design. The accuracy of the trained model, when evaluated on test data, was determined to be 91.07%. A set of nine different Yoga Moves were incorporated as the available actions in the demo due to its clearly defined joint positions. Following the training phase, the model was executed in parallel with the virtual reality (VR) system. The results are real-time transferred to the Unity application, enhancing the interactive experience within the VR environment.

To enhance the immersion of the real-world player, we introduced the use of Shimmer PPG Sensor, a heart rate monitoring device, to capture the heartbeat data of the VR player. The data collected by the Shimmer PPG sensor is used to calculate the heart rate of the VR player. The heart rate is used to control the tempo of the drumbeat, thereby allowing the rhythm of the drumbeat to dynamically adapt to the physiological state of the VR player. Each heartbeat corresponds to an eighth note in the musical composition, and each bar of the drumbeat has four quarter notes. Therefore, the variability of the heartbeat is transformed into the variability of drumbeat. The drumbeat is generated by MusicVAE, a machine learning model introduced by Roberts et al. [2018]. This model allows the use of basic heartbeat rhythm as seeds to generate complex drumbeat, and output MIDI files for real-time playback using audio samples from drum kits. With this approach, we create an audio experience which conveys the tension and excitement felt by the VR player.

3 DEMONSTRATION

At SIGGRAPH ASIA 2023 XR, we will demonstrate asymmetric collaboration between a virtual-world player and a real-world player.

Firstly, the VR player will be asked to wear an HMD to ensure that the VR environment is functioning properly. Then, the real-world player will be positioned approximately 2 meter away from the camera to ensure the whole body can be captured. When the two players are prepared, the VR player could initiate the game. Inside the VR environment, the player will encounter specific action pictures that they need to describe to the real-world player. The player in the real world needs to perform the described action in front of the camera and hold still for 3 seconds. The VR player will have a live view of the real-world player's movements through the camera feed, allowing them to provide guidance and adjustments if necessary. The intensity of the drum beat in the background music played to the real-world player will vary according to the heart rate of the VR player. The game consists of a total of nine different poses. If the real-world player successfully performs each pose accurately, the players will win the game. However, if the real-world player takes too long time or fails to perform the poses correctly, the VR player will face the consequence of virtual water flooding their environment, resulting in their loss.

4 CONCLUSION

In this demonstration, we present an immersive asymmetric collaboration experience that combines precise pose detection and emotion transmission. This system explores the application of pose detection in asymmetric cooperation, along with a novel approach of conveying emotions from a VR player to a real-world player through the drumbeat controlled by the HRV.

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